Lecture Notes on Sets, Counting, and Binomial Coefficients

N-20221205

§1.1 Summation and product notation

This note is a lecture-style introduction to sets and elementary combinatorics. It begins with notation. If $m \le n$, then

$$\sum_{k=m}^{n} a_k = a_m + a_{m+1} + \cdots + a_n,$$

and

$$\prod_{k=m}^{n} a_k = a_m a_{m+1} \cdots a_n.$$

Nested sums are allowed. For example,

$$\sum_{j=1}^{n} \sum_{k=1}^{j} a_k = a_1 + (a_1 + a_2) + \cdots + (a_1 + a_2 + \cdots + a_n).$$

The notation is not merely shorthand. It teaches us to control indices, which is essential in combinatorics and analysis.

§1.2 De Morgan's laws

For subsets A, B of a universe U, De Morgan's laws say

$$\overline{A \cup B} = \overline{A} \cap \overline{B},$$

and

$$\overline{A \cap B} = \overline{A} \cup \overline{B}.$$

The proof is an element-chasing proof. For the first identity:

$$\begin{aligned} x \in \overline{A \cup B} &\iff x \notin A \cup B \\ &\iff x \notin A \text{ and } x \notin B \\ &\iff x \in \overline{A} \cap \overline{B}. \end{aligned}$$

The handwritten margin proves the second identity in the same style. The important habit is to translate set operations into logical words: union means "or", intersection means "and", complement means "not."

§1.3 Inclusion–exclusion

For finite sets,

$$|A \cup B| = |A| + |B| - |A \cap B|.$$

For three sets,

$$\begin{aligned}|A \cup B \cup C| = |A| &+ |B| + |C| \\ &- |A \cap B| - |A \cap C| - |B \cap C| \\ &+ |A \cap B \cap C|.\end{aligned}$$

The note derives the three-set formula by applying the two-set formula to

$$A \cup (B \cup C).$$

The overcounting pattern is worth remembering: count singles, subtract pairwise overlaps, add triple overlaps back. Higher versions continue with alternating signs.

§1.4 Pigeonhole principle

The pigeonhole principle states: if $n+1$ objects are placed into n boxes, then at least one box contains at least two objects. This is almost obvious, but it is extremely powerful.

The useful mental form is contrapositive: if every box contains at most one object, then n boxes contain at most n objects. Whenever a problem asks us to prove two objects share a property, try to define boxes by that property.

§1.5 Addition and multiplication principles

The addition principle says: if a task can be completed in disjoint ways, with m_k possibilities in case k, then the total number is

$$\sum_k m_k.$$

The note's example is choosing a route from home to school: if the alternatives are parallel choices, add them.

The multiplication principle says: if a task is completed in stages, with m_k choices at stage k, then the total number is

$$\prod_k m_k.$$

This is sequential choice. The key distinction is:

$$\text{cases} \Rightarrow \text{add}, \qquad \text{stages} \Rightarrow \text{multiply}.$$

Many counting mistakes come from confusing these two.

§1.6 Permutations

To arrange n distinct objects without repetition, there are

$$n! = n(n-1)(n-2)\cdots 2 \cdot 1$$

ways. If we choose and arrange m objects from n, the number is

$$A_n^m = n(n-1)\cdots(n-m+1) = \frac{n!}{(n-m)!}.$$

The multiplication principle explains the formula: first choose the first object, then the second, and so on, with one fewer option each time.

§1.7 Combinations

If order does not matter, we divide by the number of internal arrangements. Thus the number of m-element subsets of an n-element set is

$$\binom{n}{m} = \frac{n!}{m!(n-m)!}.$$

The note remarks that different exams may write this as C_n^m or $\binom{n}{m}$. The mathematical object is the same.

The symmetry

$$\binom{n}{m} = \binom{n}{n-m}$$

has a simple interpretation: choosing the m elements included is the same as choosing the $n-m$ elements excluded.

§1.8 Pascal identity

The note asks the reader to verify

$$\binom{n}{m} = \binom{n-1}{m} + \binom{n-1}{m-1}.$$

A combinatorial proof: count m-subsets of an n-element set according to whether they contain a distinguished element x. If they do not contain x, choose all m elements from the remaining $n-1$. If they do contain x, choose the remaining $m-1$ elements from the remaining $n-1$.

This proof is better than algebraic manipulation because it explains why the identity exists.

§1.9 Binomial theorem

The binomial theorem is

$$(a+b)^n = \sum_{j=0}^{n} \binom{n}{j} a^{n-j} b^j.$$

The note suggests two proofs. The combinatorial proof expands

$$(a+b)(a+b)\cdots(a+b).$$

To get $a^{n-j}b^j$, choose exactly j of the n brackets to contribute b. This can be done in $\binom{n}{j}$ ways.

The induction proof is longer but mechanical. Both are useful, but the combinatorial proof reveals the meaning of the coefficients.

§1.10 Counting subsets

If a set S has n elements, then it has

$$2^n$$

subsets. One proof is to count by size:

$$\sum_{m=0}^{n} \binom{n}{m} = (1+1)^n = 2^n.$$

Another proof is binary choice: each element is either included or excluded.

The second proof is usually the best mental model. A subset is a yes/no decision for each element.

§1.11 Examples and exercises

The later pages contain examples involving set-builder notation, floor functions, integer lattice points, and set operations. The method is always to translate definitions into simpler conditions.

For instance, if

$$A = \{x : x^2 - 6 < 0\},$$

then

$$-\sqrt{6} < x < \sqrt{6}.$$

If a set uses the floor function $[x]$, split the real line into intervals

$$k \le x < k+1.$$

If a set contains integer pairs satisfying

$$x^2 + y^2 \le 1,$$

then draw the tiny lattice disk and count directly.

The note also has exercises asking whether certain sets intersect a circle or line family. The principle is again translation: set notation becomes equations, equations become geometry.

§1.12 Indicator functions as a unifying proof method

Many counting identities become transparent with indicator functions. For a set A, define

$$1_A(x) = \begin{cases} 1, & x \in A, \\ 0, & x \notin A. \end{cases}$$

Then

$$|A| = \sum_x 1_A(x)$$

when the universe is finite. De Morgan's laws, inclusion–exclusion, and complement counting can all be proved by checking the identity pointwise for each element x.

For two sets,

$$1_{A\cup B} = 1_A + 1_B - 1_{A\cap B}.$$

Summing over the universe gives

$$|A \cup B| = |A| + |B| - |A \cap B|.$$

For many sets, the same pointwise idea gives the full inclusion–exclusion formula.

§1.13 Combinatorial identities by double counting

The binomial identity

$$\binom{n}{r} = \binom{n-1}{r} + \binom{n-1}{r-1}$$

can be proved by deciding whether a distinguished element is chosen. If it is not chosen, choose all r elements from the remaining $n-1$. If it is chosen, choose $r-1$ more.

This is the basic style of double counting: count the same collection in two ways. It is more robust than algebraic manipulation because it explains why the identity is true.

§1.14 Conceptual synthesis

This lecture is not just "sets and counting." It introduces three ways of thinking that remain important everywhere: logical translation, structural counting, and representation switching. De Morgan turns set algebra into propositional logic; inclusion–exclusion counts by correcting overcounting; binomial coefficients count subsets and coefficients at once. These are the first signs that combinatorics is both discrete and algebraic.

§1.15 Indicator functions and inclusion–exclusion

For a finite universe U, define

$$\mathbf{1}_A(x) = \begin{cases} 1, & x \in A, \\ 0, & x \notin A. \end{cases}$$

Then

$$|A| = \sum_{x \in U} \mathbf{1}_A(x).$$

The identity

$$\mathbf{1}_{A \cup B} = \mathbf{1}_A + \mathbf{1}_B - \mathbf{1}_A \mathbf{1}_B$$

proves inclusion–exclusion after summing. For many sets, the product

$$\prod_{i=1}^{n} (1 - \mathbf{1}_{A_i})$$

encodes the complement of the union, and expanding it gives the full alternating formula.

§1.16 Inclusion–exclusion as Möbius inversion

On the Boolean lattice $\mathcal{P}([n])$, the zeta transform sends exact data to cumulative data:

$$g(S) = \sum_{T \subseteq S} f(T).$$

Möbius inversion recovers f:

$$f(S) = \sum_{T \subseteq S} (-1)^{|S|-|T|} g(T).$$

Inclusion–exclusion is precisely this inversion applied to membership in intersections of events.

This explains the alternating signs structurally. They are the inverse of summing over subsets.

§1.17 Double counting as proof by projection

A double-counting proof counts the same finite set of pairs in two projections. For example, the identity

$$\sum_{k=0}^{n} k\binom{n}{k} = n2^{n-1}$$

counts pairs (S, s) where $S \subseteq [n]$ and $s \in S$. Counting by S gives the left side; counting by s gives the right side.

This viewpoint is powerful because it forces one to name the object being counted. Once the object is named, the identity is no longer mysterious.

§1.18 Generating functions

Ordinary generating functions package a sequence into

$$A(x) = \sum_{n\geq 0} a_n x^n.$$

The coefficient extraction notation

$$[x^n]A(x) = a_n$$

turns algebra into counting. For example,

$$(1+x)^n = \sum_{k=0}^{n} \binom{n}{k} x^k$$

records subset counts by size.

Products of generating functions correspond to splitting an object into parts. This is the algebraic form of the multiplication principle.

§1.19 Pigeonhole principle and the probabilistic method

The pigeonhole principle is an averaging statement. If N objects are placed into r boxes, some box contains at least $\lceil N/r \rceil$ objects. The probabilistic method generalizes this: to prove that an object with a desired property exists, choose a random object and show that the probability of success is positive.

For instance, if the expected number of bad events is less than 1, then there must be a choice with no bad events. This is a research-level continuation of the same counting intuition: average behavior forces existence.

§1.20 Asymptotics of binomial coefficients

Pascal's triangle has a limiting shape. Stirling's formula gives

$$\binom{n}{\lfloor n/2 \rfloor} \sim \frac{2^n}{\sqrt{\pi n/2}}.$$

Probabilistically, the binomial coefficients are the distribution of a sum of independent Bernoulli variables. After centering and scaling, the central limit theorem gives a Gaussian approximation.

So the finite identity $\sum_k \binom{n}{k} = 2^n$ is the exact count, while asymptotic analysis explains the shape of the whole row for large n.

§1.21 Combinatorial species

A combinatorial species assigns to each finite label set U a set $F[U]$ of structures, compatibly with relabelling bijections. The species of subsets sends U to $\mathcal{P}(U)$. A bijection of label sets transports subsets.

Species theory records not just how many objects exist, but how constructions behave under relabelling. Addition, product, and composition of species categorify the elementary counting principles.

§1.22 Counting as inversion, projection, and generation

The lecture's elementary tools have precise advanced continuations. Set logic becomes Boolean algebra, inclusion–exclusion becomes Möbius inversion, double counting becomes projection of incidence sets, generating functions encode sequences, pigeonhole becomes averaging and probabilistic existence, and binomial coefficients connect exact counting with asymptotic probability.